

HIGH LEVEL DESIGN

NXT Automated Signal and Trading System

Thuộc tính	Giá trị
Phiên bản	v0.1 — Desktop-first architecture
Ngày	15/07/2026
Nguồn yêu cầu	BRD NXT Automated Signal and Trading System v0.2
Strategy baseline	NXT v3.5 USD-M BTCUSDT/BNBUSDT/SOLUSDT 1D; promoted rulebook 2026-07-14
Runtime MVP	Windows desktop: local app + Windows Task Scheduler + local durable store
Execution boundary	Binance USD-M Futures testnet first; live only after explicit approval and go-live controls

Architecture decision

MVP is a desktop-first modular monolith. Logical modules are independently testable, but are deployed locally together to minimize operational complexity. Cloud hosting is deliberately out of MVP scope and remains a later readiness decision.

1. Purpose and scope

This HLD defines the target architecture for a human-in-the-loop NXT system: it scans a promoted NXT ruleset, produces an approved order proposal, executes only within policy, monitors real exchange state, and records an auditable lifecycle.

In scope	Out of scope / deferred
Desktop scan, rule evaluation, approval workflow, USD-M testnet gateway, order protection, reconciliation, audit, Telegram/local UI.	Cloud runtime, autonomous rule promotion, HFT/intraday, multi-exchange routing, withdrawal automation, unattended live entry approval.

2. Architecture drivers and principles

Driver	Architectural response
Human approval	Proposal and Decision are durable records. Execution accepts only a non-expired approved proposal hash.
NXT latest changes	RuleVersion is immutable with parameters, artifact hash, source contract/session, and promotion timestamp. Existing positions retain their entry rule version.
USD-M execution risk	Risk engine sizes before leverage; isolated margin, exchange filters, liquidation buffer and portfolio caps are pre-trade gates.

Driver	Architectural response
Ambiguous exchange outcomes	Stable clientOrderId, query-before-retry, idempotency ledger and Binance reconciliation prevent duplicate logical orders.
Desktop-first reliability	Task Scheduler invokes a controlled service command; local health status, StartWhenAvailable, checkpoint recovery and operator alerting cover host interruptions.
Auditability	Append-only AuditEvent records actor, source, before/after state, correlation ID and external order IDs.

3. Logical architecture

NXT desktop-first logical architecture

Windows local runtime | Binance USD-M is the execution system of record

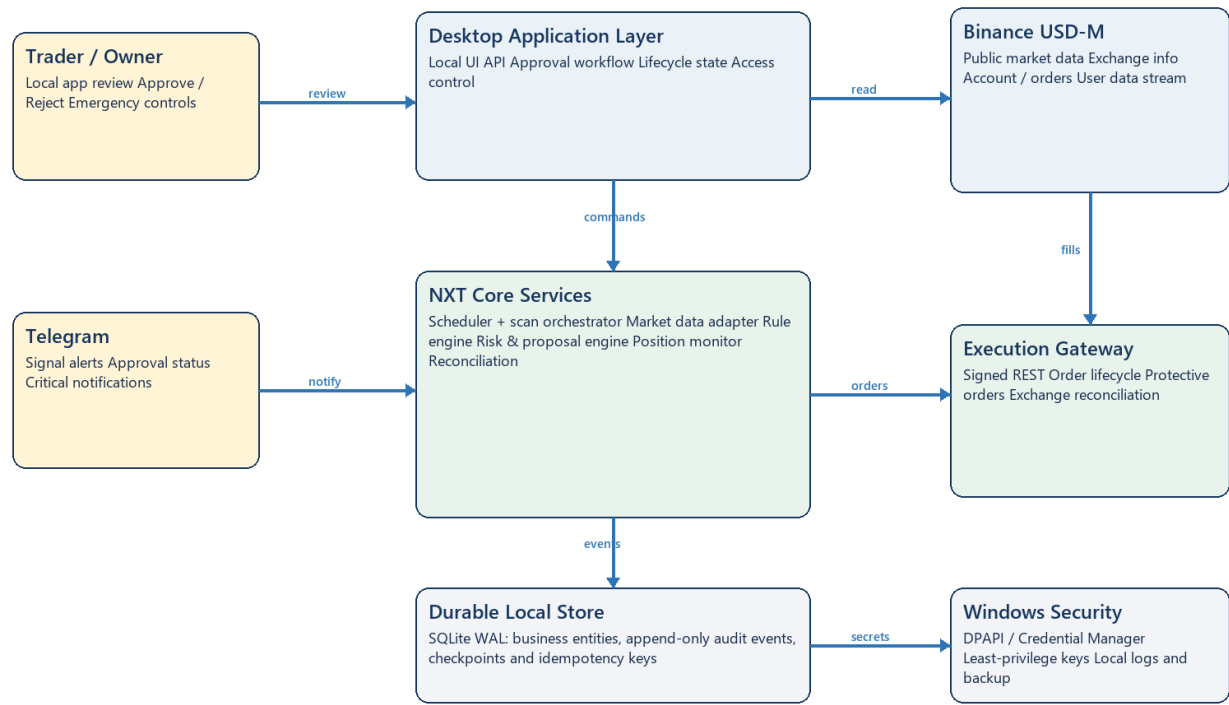


Figure 1. Desktop-first logical architecture

Layer	Components	Responsibility
Experience	Local App; Telegram notifier	Review proposals, display lifecycle/audit, capture Approve/Reject and provide alerts. Telegram is notification-first; privileged decisions remain policy-controlled.
Application	Workflow service; lifecycle manager; policy evaluator	Owns state transitions, approval expiry, idempotency and exception-to-Needs Review behavior.
Strategy	Market data adapter; NXT rule engine; RuleVersion registry	Reads closed USD-M candles, computes indicators, evaluates E/P/G rules, stores pass/fail rationale and signal snapshot.

Layer	Components	Responsibility
Trading	Risk/proposal engine; execution gateway; position monitor	Calculates quantity/margin, validates filters, submits approved orders, maintains protective orders and reconciles exchange events.
Data/control	SQLite WAL store; audit log; settings/secrets; backup exporter	Durable local source for business state; Binance remains source of truth for exchange order/fill/position facts.
External	Binance USD-M APIs/streams; Telegram API	Market data, exchange status, trading/account events, delivery of notifications.

4. Desktop deployment topology

Node	Processes / stores	Operational responsibility
Windows desktop host	NXT local app/service; scheduled scan command; SQLite database; encrypted secrets; local log and backup folder.	Primary MVP runtime. Service must start after login/boot according to Task Scheduler configuration and publish health status.
Binance USD-M	Public market data; exchange information; signed REST; user data stream.	Execution system of record. All uncertain outcomes reconcile from Binance before another action.
Telegram	Bot API outbound notifications and optional review deep-link.	Secondary interaction channel. Notification failure must not lose persisted signal/proposal state.

Desktop boundary

No inbound public internet service is required for MVP. The local app binds to loopback by default. Remote access, public endpoints and cloud workers are not part of this HLD's deployment baseline.

5. Core workflows

5.1 Daily scan and proposal

Step	Flow and control
1. Schedule	Windows Task Scheduler starts the scan after USD-M 1D candle close (default 07:10 ICT). A ScanRun receives a correlation ID.
2. Data quality	Market Data Adapter verifies symbol, source, closed candle, freshness and expected session. Stale/incomplete data produces no new signal and raises an operator alert.
3. Rule evaluation	NXT Rule Engine loads the promoted immutable RuleVersion and creates a Signal snapshot only after all E/P/G conditions pass, including G-02.
4. Proposal	Risk Engine reads account snapshot, open-risk cap, exchange filters and policy; it computes quantity, notional, isolated margin requirement, leverage constraint, SL/TP1 and expiry.

Step	Flow and control
5. Persist then notify	Workflow persists Signal and OrderProposal before Telegram/local UI notification. Signal deduplication uses versioned deterministic keys.
6. Decision	Trader approves/rejects the exact proposal hash. A changed price, expired proposal, changed policy or changed levels invalidates approval.

5.2 Execution and protection

Step	Flow and control
1. Pre-trade revalidation	Execution Gateway rechecks API permission, server time, account/margin, position state, leverage/margin mode, price deviation, filters, risk limits and idempotency.
2. Entry	Gateway writes clientOrderId before submit. A timeout or 5xx is an unknown outcome: query order and user stream before retrying.
3. Fill handling	Actual fills update Position and ExchangeOrder. Partial fill quantity becomes the maximum size for protective orders.
4. Protection	Create reduce-only SL and TP1 immediately. Rejection creates Critical state, blocks new entries and follows owner-approved fail-safe policy.
5. Exit lifecycle	Monitor detects TP1, Early-BE, stop and SSL runner action. Action is a recommendation or automated only at the approved automation level.

5.3 Monitoring and reconciliation

- User data stream is the low-latency feed for orders, fills and account/position changes; REST reconciliation repairs gaps after disconnect or restart.
- Binance is the source of truth for actual orders, fills and positions. Internal state is a reconciled projection, not a substitute for exchange facts.
- Every daily close runs a position recommendation using the entry RuleVersion or an explicit migration record. A new latest version cannot silently change an open position's policy.
- Any mismatch, ambiguous response, stale data or rejected protective order becomes Needs Review and suppresses dependent automation.

6. Data architecture

Entity	Key contents	Control
RuleVersion	version, parameters, USD-M contract/session, system document hash, promotedAt	Immutable and retained indefinitely.
ScanRun	runId, candle date, symbols, source/freshness, status, errors, duration	Unique run ID; operational retention at least two years.
Signal	deterministic signalId, RuleVersion, indicator snapshot, guard rationale, levels	Immutable once detected; deduplicated by source + rule version.

Entity	Key contents	Control
OrderProposal / Decision	proposal hash, quantity, risk, expiry, decision actor/channel/time	No in-place change after approval. New values require new proposal/decision.
ExchangeOrder / Position	clientOrderId, Binance order ID, fills, protective order IDs, actual quantity/lifecycle	Reconciled with exchange events and REST snapshots.
Recommendation / AuditEvent	action rationale, automation level, status; actor, before/after, correlation ID	Append-only; supports investigation and weekly audit.

Local storage selection

Proposed MVP store: SQLite in WAL mode with transactional writes, periodic encrypted backup/export and a migration layer. This meets the single-host desktop scope while preserving a future move to a managed database if cloud is approved.

7. Interfaces and integration boundaries

Interface	Direction	Purpose and resilience
USD-M market data	Outbound read	Fetch candles and exchange info. Validate closed bar and freshness; cache may support display only, never new signal generation when stale.
USD-M trading/account	Outbound signed REST	Query account/order status; submit/cancel/replace approved orders; use clientOrderId and query-before-retry.
USD-M user data stream	Inbound event	Receive account/order/fill events; reconnect and backfill via REST when sequence or connectivity is uncertain.
Telegram Bot API	Outbound notification	Signal, decision, fill and critical alerts. Retry delivery independently of business state persistence.
Local app API	Loopback only	UI reads business state and records permitted user actions. No public listener in MVP.

8. Security, risk and resilience

Concern	Design control
Secrets	Store API credentials in Windows Credential Manager/DPAPI-backed storage; never in source, SQLite, exports or logs. Keys must lack withdrawal permission.
Environment separation	Testnet and live use distinct configuration, credentials, clientOrderId namespace, database profile and visible UI banner.
Leverage and liquidation	Risk budget determines quantity first. Leverage is a margin-efficiency setting, capped by policy; pre-trade validation verifies isolated margin and liquidation buffer before approval/execution.
Failure handling	Unknown exchange outcome, stale data, protective failure or reconciliation mismatch transitions to Needs Review/Critical rather than inferring success.

Concern	Design control
Kill switch	Pause blocks new entry while monitoring/protection continues. Kill switch blocks automated action except an owner-approved emergency policy; all changes are audited.
Desktop continuity	Task Scheduler StartWhenAvailable, health check, alerting, checkpointed ScanRun and reconciliation on restart. Cloud is not the fallback in MVP.

9. NFR and requirement traceability

BRD area	HLD response	Verification evidence
FR-01 / FR-02; BR-01 to BR-11	Scheduler, data-quality gate, NXT Rule Engine and RuleVersion registry.	Golden-signal test; G-02 regression; deterministic Signal ID.
FR-03 to FR-07; BR-12 to BR-15	Risk/Proposal Engine, Approval workflow, Execution Gateway and Protective Order manager.	Testnet E2E; pre-trade checks; partial fill and protective failure tests.
FR-08 / FR-09; BR-16 / BR-17	Position monitor, user data adapter, REST reconciliation and recommendation service.	Disconnect/restart/manual intervention reconciliation evidence.
FR-10 to FR-12; NFR-02/03/05/07	Notifier, local dashboard, audit store, health monitor and operational controls.	Delivery retry logs, audit timeline, kill-switch UAT and scheduled-run status.
Desktop-first decision	Single-host modular deployment and local data store; cloud excluded from MVP boundary.	30-day desktop paper/testnet operational run.

10. Key design decisions and open decisions

Decision	Status	Rationale / remaining action
Desktop-first modular monolith	Confirmed	Lowest operational complexity for MVP; components remain separable for later cloud assessment.
Binance USD-M as technical baseline	Confirmed	Matches promoted NXT data and funding model. Owner must still confirm account eligibility/contract availability before testnet/live.
Human approval for entry	Confirmed	No live entry without an approval record matching the proposal hash and environment.
Live risk, leverage ceiling and loss caps	Open / blocking	Owner must set per-trade and portfolio policies. HLD enforces a policy interface rather than hardcoding values.
Runner exit automation level	Open / high	Recommendation-only is the safe default until owner approves an automation policy.
Cloud readiness	Deferred	Evaluate only after desktop paper/testnet pilot achieves the BRD operational-run exit criteria.

11. Phased implementation view

Phase	Architecture increment	Exit condition
1. Desktop paper	SQLite schema, RuleVersion registry, scan workflow, local UI/Telegram approval, simulated exchange adapter and audit timeline.	Golden signal, approval, duplicate and restart UAT pass.
2. Desktop USD-M testnet	Signed gateway, exchange filters, user stream, protective orders, reconciliation, risk/portfolio pre-trade gates.	30-day stable run without duplicate order or unresolved critical protection failure.
3. Desktop live assisted	Feature-gated live profile, manual entry approval, production monitoring, backup/restore drill and low-risk policy configuration.	Go-live checklist and owner approval.
4. Later assessment	Optional managed automation and cloud-readiness design only after desktop evidence supports it.	Separate decision record; no automatic migration.

12. HLD acceptance criteria

- Architecture supports every Must requirement in BRD v0.2 without requiring cloud deployment.
- Signals, proposals, decisions and exchange actions are traceable by correlation ID and immutable rule/proposal references.
- No execution path bypasses approval, pre-trade checks, idempotency or protective-order validation.
- Rule update, G-02 regression, stale-data block, restart recovery and manual Binance action have explicit handling paths.
- All open risk, leverage, margin and automation values remain policy-configured and require owner approval before live use.

Appendix A. References

- docs/BRD_NXT_Automated_Signal_Trading_System_v0.2.docx
- latest/NXT_Latest_Summary.md
- latest/NXT_Latest_NXT35_USDM_BlockShortAfterLosingLong_System_And_Indicators.docx
- latest/NXT_Latest_NXT35_USDM_BlockShortAfterLosingLong_SignalRegression.json
- app/nxt_signal_app.py and scripts/daily_nxt_signal_scan.py — current desktop scan core.